Android Recovery Developer Guide

ID: RK-KF-YF-E38

Release Version: V1.3.0

Release Date: 2024-12-12

Security Level: □Top-Secret □Secret □Internal ■Public

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Preface

Overview

This document mainly describes Rockchip Recovery System instruction.

Product Version

Chip Name	Kernel Version
RK3326/RK3399/RK356x/RK3588	4.19, 5.10

Intended Audience

This document (this guide) is mainly intended for

Technical Support Engineer

Software Development Engineer

Revision History

Version	Author	Date	Change Description
V1.0.0	Ji Dayao	2023-05-11	Initial version release
V1.1.0	Ji Dayao	2023-05-15	Updated Section 2
V1.2.0	Ji Dayao	2023-10-28	Updated Section 2.3, introducing the compilation command for Android 14 difference package
V1.3.0	Ji Dayao	2024-12-12	Updated Section 2, added GRF OTA compilation introduction

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1. Overview

This document describes the instruction for Rockchip Recovery system, which can be used on Rockchip Android >= 12 platforms.

Recovery mode refers to a mode that allows modification of internal data or the system on an Android device. In this mode, users can flash a new Android system, back up or upgrade an existing system, or perform a factory reset. This document primarily targets non-AB systems. For AB systems, please refer to the document 《Rockchip_Developer_Guide_Android_AB_System_Upgrading_CN》.

2. Compiling OTA Packages

On Rockchip Android platforms, the A/B system feature is disabled by default. To enable the A/B system, configuration is required across three aspects: the Android system, U-BOOT, and kernel dts. For detailed instructions, please refer to the document 《Rockchip_Developer_Guide_Android_AB_System_Upgrading_CN》. This document primarily focuses on non-A/B system configurations.

Rockchip supports the GRF platform starting from Android 15 (i.e., >= Android 15). For the compilation methods of OTA full packages and delta packages on the Android GRF platform, refer to 《Rockchip_Developer_Guide_Android_GRF_OTA》.

2.1 OTA Overview

OTA (Over The Air) update is a standard software update method provided by the Android system. It is powerful and offers full update (full package), incremental update mode (delta package), and supports both local and network-based updates.

2.2 Generating the Complete Package

OTA (Over The Air) update is a standard software update method provided by the Android system. It is powerful and supports both full-package update (complete package) and incremental update mode (delta package), capable of performing updates via local or network.

Compilation commands are as follows:

```
Non-AB mode:
make installclean && make -j16 && make dist -j16 && ./mkimage.sh ota
AB mode:
make installclean && make -j16 && make dist -j16 && ./mkimage_ab.sh ota
```

Alternatively, as follows:

```
Non-AB mode:
build.sh -AUCKuop
AB mode:
build.sh -ABUCKuop
```

In the out/target/product/rkxxxx/ directory, an OTA full package named rkxxxx-ota-eng.root.zip will be generated. This can be renamed to update.zip and then copied to an SD card or internal flash storage for system update.

Additionally, various Android platforms can be compiled using the build.sh script. When using build.sh, the -o option must be included, for example: build.sh –AUCKuop.

Firmware releases must use ./mkimage.sh ota to package boot and kernel images together. No need to flash the kernel separately. If production firmware images are separate, this will impact subsequent delta package updates unless delta updates are not required.

If using ./build.sh to package update.img, you must use ./build.sh -ou, and cannot simply use ./build.sh -u

2.3 Generating Delta Package

OTA delta packages contain only differential content, with smaller package sizes, and are primarily used for OTA online upgrades, and can also be used for T-card local upgrades. The creation of OTA delta packages requires a special manual compilation.

Steps as follows:

1. First, release the firmware version v1 and generate the full package for version v1. The compilation command is as follows:

```
Non-A/B:
make installclean && make -j16 && make dist -j16 && ./mkimage.sh ota
A/B:
make installclean && make -j16 && make dist -j16 && ./mkimage_ab.sh ota
```

or

```
Non-AB mode:
build.sh -AUCKuop
AB mode:
build.sh -ABUCKuop
```

Save out/target/product/rkxxxx/obj/PACKAGING/target_files_intermediates/rkxxx-target_files-eng.xxx.zip as rkxxxx-target_files-v1.zip as the base material package for the v1 version. Simultaneously flash the firmware under rockdev from the first-step compilation onto the device (Note: This is critically important).

3. Modify kernel code or Android code, release v2 version firmware, and generate complete package for v2 version. Compilation commands are as follows:

```
   make installclean && make -j16 && make otapackage -j16
```

or

```
Non-AB mode:
build.sh -AUCKuop
AB mode:
build.sh -ABUCKuop
```

4. Save

out/target/product/rkxxxx/obj/PACKAGING/target_files_intermediates/rkxxx-target_files-eng.xxx.zip is renamed to rkxxxx-target_files-v2.zip, serving as the base material package for version v2.

5. Generate the v1-v2 delta upgrade package. Refer to the following commands:

```
   ./build/tools/releasetools/ota_from_target_files -v -i ./rockdev/v1/rk3588_t-
target_files-eng.jdy.zip --block -p ./out/host/linux-x86  ./rockdev/v2/rk3588_t-
target_files-eng.jdy.zip  ./rockdev/v1v2.zip
```

For Android 13, Python 3 must be used. Refer to the following commands:

```
   python3 ./build/tools/releasetools/ota_from_target_files -v -i
./rockdev/v1/rk3588_t-target_files-eng.jdy.zip --block -p ./out/host/linux-x86
./rockdev/v2/rk3588_t-target_files-eng.jdy.zip  ./rockdev/v1v2.zip
```

For >=Android 14, refer to the following commands:

```
   out/host/linux-x86/bin/ota_from_target_files -v -i ./rockdev/V1/rk3588_u-
target_files-eng.jdy.zip  --block -p ./out/host/linux-x86
./rockdev/V2/rk3588_u-target_files-eng.jdy.zip  ./rockdev/v1v2.zip
```

Description: The command format to generate Delta Package:

ota_from_target_files

-v -i the previous target file for comparison

--block Use block mode for OTA update

-p host compilation environment

-k packaging key, the default key path for Android 13 is device/rockchip/common/security/testkey, while for other Android platforms the default key path is build/make/target/product/security/testkey.

The subsequent target file for comparison

Generated OTA Delta Package

2.4 Notes

The full package and resource package are different. As can be known from the method of generating delta packages, the delta package is generated by differencing two versions of resource packages. Therefore, each time a firmware version is released, the resource package under the obj directory, along with the full package for upgrades and the various images generated by executing ./mkimage.sh ota must be preserved.

3. rkLoader Upgrade

Under normal circumstances, an OTA update does not require upgrading rkloader. When an rkloader upgrade is needed, place the loader to be upgraded (i.e., the miniloader used by the burning tool for flashing) in the format of RK*Loader*.bin into the directory $(TARGET_DEVICE_DIR)/ota/loader. During the OTA packaging process, it will be included in the update package.

The specific value of $(TARGET_DEVICE_DIR) can be confirmed by executing get_build_var TARGET_DEVICE_DIR after sourcing build/envsetup.sh and running lunch XXX in the SDK root directory.

4. Directly Upgrade update.img in Recovery

In normal circumstances update.img is solely used for tool flashing. To upgrade update.img within Recovery, simply invoke the do_rk_mode_update interface in RK Recovery at appropriate positions according to the customer's own application logic. For example, when upgrading the update.img via a USB drive, before invoking the interface, ensure the USB drive is mounted first - you can manually mount it or directly invoke the following interface:

```
prksdboot->ensure_usb_mounted();
```

The reference usage method for the do_rk_mode_update interface is as follows (assuming that update.img is located on the USB drive):

```
+ auto ui = device->GetUI();
+ InstallResult result = INSTALL_SUCCESS;

+ prksdboot->sdboot_set_bSDBoot(false);
+ prksdboot->sdboot_set_bUsbBoot(true);
+ /*FileName is the path relative to the USB drive's root directory. For
example, if update.img is located in the "abc" subdirectory under the USB root,
set FileName to "/abc/update.img"*/
+ result = (InstallResult)do_rk_mode_update(FileName, prksdboot, ui);
```

5. Configuration Instruction

5.1 Log Redirection

1. Function Description: Log can be output to serial port, SD card and /cache/recovery/ three locations.
2. Opening method:

   bootable/recovery/Android.bp:

-DLogToSerial Output logs to the serial port

-DLogToCache Output logs to the /cache/recovery/ directory

-DLogToSDCard Output logs to recovery.log in the SD card

5.2 Screen Rotation

1. Function Description: The Recovery Screen can be rotated to 0°, 90°, 180°, or 270°.
2. Method of Access: Modify the device/rockchip/XXXX/BoardConfig.mk file:

```
TARGET_RECOVERY_DEFAULT_ROTATION := ROTATION_NONE No rotation
TARGET_RECOVERY_DEFAULT_ROTATION := ROTATION_LEFT Rotation 270
TARGET_RECOVERY_DEFAULT_ROTATION := ROTATION_RIGHT Rotation 90
TARGET_RECOVERY_DEFAULT_ROTATION := ROTATION_DOWN Rotation 180
```

5.3 File Saving

Save the file to /cache/recovery/Recovery_*, formatting will not erase this file.

5.4 NTFS Formatted USB Drives/SD Cards Upgrade Support

NTFS file system currently has intellectual property issues and is not supported by default. If NTFS is to be used, ensure relevant rights holder authorization is obtained prior to utilization.

The kernel does not natively support mounting NTFS-formatted USB drives, hence NTFS is unsupported in recovery. To enable NTFS support, simply enable the relevant configuration within the kernel.

6. Upgrade Steps

For validating the RK platform, RK provides by default the upgrade client RKUpdateService for validation. The steps are as follows. Note that if it is an AB system, please refer to 《Rockchip_Developer_Guide_Android_AB_System_Upgrading_CN》 for validation.

1. Refer to "2. Compile OTA Package", generate the OTA package and rename it to update.zip.
2. Copy update.zip to the root directory of USB, SD card, or the /data/media/0/ directory.
3. The system will automatically detect the update package and display the update dialog box. If automatic detection fails, it is recommended to replug the USB cable or directly restart the device.
4. Automatic Reboot Upgrade: Reboot into the Recovery system to automatically complete the OTA package upgrade. Wait for the upgrade to finish, after which it will automatically reboot to the Android main interface.

7. Upgrade Card

7.1 SD Upgrade Card

The SDDiskTool can be used to create an upgrade card from the production firmware 'update.img'. When using the SDDiskTool, select 'Firmware Upgrade' in the Function Mode, then choose the corresponding update firmware 'update.img', and finally click 'Start Creation' to wait for the card creation to complete.

Once the upgrade card is completed, insert it into the device's corresponding hardware interface and then power on the device.

For the AB system, before card creation, it is necessary to configure the SDDiskTool tool's config by setting UPGRADE_PROGRAM_PARTITION=boot_b, then re-open SDDiskTool to proceed with card creation.

7.2 USB Upgrade Disk

If you want to use USB to create an upgrade disk, you first need to enable the CONFIG_ROCKCHIP_USB_BOOT configuration in u-boot, then recompile the u-boot, flash the u-boot with this configuration into the device, and finally use the fully compiled update.img to create the upgrade disk.

Taking rk3588 as an example, the reference configuration for u-boot is as follows:

```
diff --git a/configs/rk3588_defconfig b/configs/rk3588_defconfig
index 67a8c357f0..be47518735 100644
--- a/configs/rk3588_defconfig
+++ b/configs/rk3588_defconfig
@@ -234,3 +234,4 @@ CONFIG_RK_AVB_LIBAVB_USER=y
 CONFIG_OPTEE_CLIENT=y
 CONFIG_OPTEE_V2=y
 CONFIG_OPTEE_ALWAYS_USE_SECURITY_PARTITION=y
+CONFIG_ROCKCHIP_USB_BOOT=y
```

The SDDiskTool can be used to create an upgrade disk from the production firmware update.img. When using the SDDiskTool, select 'Firmware Upgrade' in the function mode, then choose the corresponding upgrade firmware update.img, finally click 'Start Creation', and wait for the disk creation to complete.

After completing the upgrade disk creation, insert the device into the corresponding hardware interface and power on the device.

For the AB system, before the disk creation, you must set UPGRADE_PROGRAM_PARTITION=boot_b in the config of the SDDiskTool, then re-open the SDDiskTool for the disk creation.